NMEC595 - Research Methodology

Lecture 8: Data Analysis - Tabulation, Central Tendency & Dispersion

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Winter Semester 2026

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Where We Are in the Research Journey

Previous Lecture (Lecture 6):

We learned to PROCESS raw data:

- ✓ **EDITING** - detected and corrected errors
- ✓ **CODING** - standardized diverse responses
- ✓ **CLASSIFICATION** - organized into meaningful groups
- ✓ **TABULATION** - presented in professional tables

Today's Challenge:

We have organized, tabulated data. But **the data still holds no meaning**.

Key Questions:

- What is the "typical" value?
- How spread out is the data?
- Is the process consistent or variable?
- Can we make predictions?

Today's focus: STATISTICAL ANALYSIS

A Fundamental Question

A manufacturing plant produces steel rods. Quality control tests 50 rods for diameter:

Results range from 9.8 mm to 10.2 mm

CEO asks manager: "What diameter should we put on the product label?"

Manager's poor answer: "Somewhere between 9.8 and 10.2 mm"

What manager needs to answer:

What is the TYPICAL or AVERAGE diameter?

This is the essence of Central Tendency - finding the "center" of a dataset

Three ways to measure "center":

1. Mean - arithmetic average (most common)
2. Median - middle value (robust to outliers)
3. Mode - most frequent value (qualitative data)

The Mean: Arithmetic Average

Definition: Sum of all values divided by the count

Formula:

$$\bar{X} = \frac{\sum X}{n} = \frac{X_1 + X_2 + ... + X_n}{n}$$

Real Example - Steel Rod Diameter:

10 rods measured: 10.05, 10.08, 10.02, 10.10, 10.00, 10.07, 10.04, 10.06, 10.03, 10.05 mm

$$\bar{X} = \frac{10.05 + 10.08 + ... + 10.05}{10} = \frac{100.50}{10} = 10.05 \text{ mm}$$

Now we can say: "Average rod diameter is 10.05 mm"

The Mean: Arithmetic Average

Why use Mean?

- **Uses all data points** - no information lost
- **Basis for further analysis** - enables variance, standard deviation
- **Matches intuition** - what most people call "average"

Caution: Mean is sensitive to extreme values!

Example: If one rod was 15.00 mm (manufacturing error):

$$\bar{X} = \frac{100.50 + 4.95}{11} = 9.59 \text{ mm}$$

Mean drops from 10.05 to 9.59 mm - **misleading!**

The Median: Robust Middle Value

Definition: The value that falls exactly in the middle when data ordered

Procedure:

1. Arrange all values in order (ascending or descending)
2. Find middle position: Position $= \frac{n+1}{2}$
3. Value at that position is the median

Example 1 - Odd Number of Values:

9 rods (ordered): 10.00, 10.02, 10.03, 10.04, 10.05, 10.06, 10.07, 10.08, 10.10 mm

Position: $\frac{9+1}{2} = 5$

Median = 10.05 mm (5th value)

The Median: Robust Middle Value

Example 2 - Even Number of Values:

Same 9 rods + one more (10.05 mm):

Ordered: 10.00, 10.02, 10.03, 10.04, 10.05, **10.05**, 10.06, 10.07, 10.08, 10.10

Position: $\frac{10+1}{2} = 5.5 \rightarrow$ Average 5th and 6th values

Median = $\frac{10.05+10.05}{2} = 10.05$ mm

The Powerful Advantage of Median:

Same extreme value example (rod at 15.00 mm):

Ordered: 10.00, 10.02, 10.03, 10.04, 10.05, 10.05, 10.06, 10.07, 10.08, 10.10, **15.00**

Position: $\frac{11+1}{2} = 6$

Median = 10.05 mm (6th value)

Median unchanged! Not fooled by the outlier.

The Mode: Most Frequent Value

Definition: The value that appears most frequently in the dataset

Example 1 - Quantitative Data:
Hardness measurements (HV): 450, 455, 455, 455, 460, 465, 470, 470, 475
Mode = 455 HV (appears 3 times, more than any other)

Example 2 - Qualitative Data:
Material preference survey (30 engineers asked: "Preferred surface finish?"):

- Polished: 12 engineers
- Ground: 8 engineers
- Machined: 5 engineers
- Raw: 5 engineers

Mode = Polished (12 votes, most frequent preference)

The Mode: Most Frequent Value

Unique Advantage of Mode:

- **Works with qualitative data** - the ONLY measure for categorical variables
- **Represents actual occurrence** - no mathematical manipulation
- **Intuitive** - everyone understands "most common"

Limitations:

- **May not exist** - if all values appear equally
- **May not be unique** - multiple modes possible (bimodal, multimodal)
- **Ignores numerical values** - doesn't use magnitude information

Mean vs. Median vs. Mode: Decision Guide

When should you report each?

Situation	Mean	Median	Mode	Report
Normal data				Mean
No outliers				Mean
Few extremes				Both
Many outliers				Median
Qualitative data				Mode
Bimodal/skewed				All three

Practical Recommendation - Report BOTH Mean and Median:

- If Mean $\approx$ Median $\rightarrow$ data is symmetric
- If Mean » Median $\rightarrow$ **right skew** (outliers on high end)
- If Mean « Median $\rightarrow$ **left skew** (outliers on low end)

Mean vs. Median vs. Mode: Decision Guide

Real Example - Fatigue Life Testing:

Mean = 150,000 cycles; Median = 100,000 cycles

Interpretation:
A few cables lasted very long (right skew). For conservative design, use Median (100,000 cycles) as the baseline.

MCQ: Adding a Constant (Curve)

If we add a constant c to every score in a dataset, what happens to the mean and median?

A) Mean stays the same; median increases by c
B) Mean increases by c; median increases by c
C) Mean increases by c; median stays the same
D) Mean and median both multiply by c

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Correct: B

MCQ: Multiplying by a Constant (Scaling)

If every observation in a dataset is multiplied by a constant c, what happens to the mean and median?

A) Mean and median both increase by c
B) Mean multiplies by c; median stays the same
C) Mean and median both multiply by c
D) Mean and median are unchanged

MCQ: Multiplying by a Constant (Scaling)

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D) Mean and median are unchanged
Correct: C

Note: Variance does not follow the same rule.

Beyond Central Tendencies

Manufacturing Quality Crisis

Two suppliers provide steel plates:

Supplier A: Thickness measurements: 5.0, 5.0, 5.0, 5.0, 5.0 mm → **Mean = 5.0 mm**

Supplier B: Thickness measurements: 4.0, 4.5, 5.0, 5.5, 6.0 mm → **Mean = 5.0 mm**

Same mean, but completely different quality!

Which supplier would you choose?

Supplier A: Perfect consistency, all exactly 5.0 mm Supplier B: Highly variable, could be 4.0–6.0 mm

For engineering, consistency is critical!
We need a measure of SPREAD or VARIABILITY
This is called DISPERSION or VARIATION

Three Basic Measures of Variability

1. **Range**
   - Difference between maximum and minimum values
2. **Interquartile Range (IQR)**
   - Spread of the middle 50% of data
3. **Standard Deviation**
   - Average distance of data from the mean

Purpose: All three describe how spread out the data are, but each gives a different perspective.

Range: Simplest Dispersion Measure

Definition: Difference between maximum and minimum values

$$\text{Range} = X_{\max} - X_{\min}$$

Example - Tensile Strength of Steel Bars:
Test results: 430, 450, 465, 480, 495, 510 MPa

$$\text{Range} = 510 - 430 = 80 \text{ MPa}$$

Interpretation: Strength varies by 80 MPa across samples.

Advantages:

Extremely simple - any engineer can calculate instantly **Intuitive** - everyone understands "spread from min to max" **Quick quality check** - identifies extreme values

Critical Disadvantages:

Only uses 2 values - ignores all data in the middle **Highly sensitive to outliers** - one extreme value ruins the measure **Not suitable for statistics** - can't use for predictions or confidence intervals

Quartiles: Dividing Data into Four Equal Parts

Quartiles are values that divide a ranked (ordered) dataset into four equal parts.

- **First Quartile (Q_1):** Value below which 25% of data fall.
- **Second Quartile (Q_2):** Median; 50% of data fall below.
- **Third Quartile (Q_3):** Value below which 75% of data fall.

Purpose: Quartiles help describe the distribution and spread of data.

Interquartile Range (IQR): Definition and Purpose

Definition: The Interquartile Range (IQR) measures statistical dispersion as the difference between the third and first quartiles.

$$\text{IQR} = Q_3 - Q_1$$

Key Point: IQR shows the range covered by the central 50% of the data and ignores extreme values (outliers).

Why use IQR?

- Less sensitive to outliers than the range.
- Focuses on the typical (middle) data spread.

Example

Dataset (increasing order):
24, 58, 61, 67, 71, 73, 76, 79, 82, 83, 85, 87, 88, 88, 92, 93, 94, 97

Quartiles split the data into four equal parts:

- $\mathbf{Q}_1$**:** First quartile (lower 25%)
- $\mathbf{Q}_2$**:** Median (middle value, 50%)
- $\mathbf{Q}_3$**:** Third quartile (lower 75%)

$n = 18$ (number of students)

Calculating Quartiles (with Interpolation)

Quartile positions (for $n = 18$):

- Q_1 position: $\dfrac{n+1}{4} = \dfrac{19}{4} = 4.75$
- Q_2 (median) position: $\dfrac{n+1}{2} = \dfrac{19}{2} = 9.5$
- Q_3 position: $\dfrac{3(n+1)}{4} = \dfrac{3 \times 19}{4} = 14.25$

Use interpolation between ranked values:

$$Q_1 = 67 \text{ (4th)} + 0.75 \times (71 - 67) = 70$$
$$Q_2 = 82 \text{ (9th)} + 0.5 \times (83 - 82) = 82.5$$
$$Q_3 = 88 \text{ (14th)} + 0.25 \times (92 - 88) = 89$$

Interquartile Range (IQR) and Interpretation

Definition: $\text{IQR} = Q_3 - Q_1$

From previous calculations:

$$\text{IQR} = 89 - 70 = 19$$

Interpretation: The central 50% of exam scores are within a span of 19 points.

The Five Number Summary

A helpful summary of a dataset is called the **five number summary**. It consists of five values:

1. The minimum
2. The lower quartile (Q_1)
3. The median (also known as Q_2)
4. The upper quartile (Q_3)
5. The maximum

These values provide a quick overview of the distribution and spread of data.

Variance

Another way to describe the spread (variability) of data is by computing the **variance**.

Definition: Variance is the average squared distance from the mean.

Population Variance:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

- μ is the population mean
- N is the population size

Sample Variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1} = \frac{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2}{n-1}$$

- $\overline{x}$ is the sample mean
- n is the sample size

Variance

Why squared deviations?

- Simple deviations may sum to a small number (positive and negative cancel)
- Squaring makes all deviations positive
- Emphasizes larger deviations more

Why squared deviations?

When calculating the sample variance or standard deviation, we estimate the population mean using the sample mean.
Dividing by $(n-1)$ (instead of n) gives the sample variance a special property—it becomes an **unbiased estimator** of the population variance.

- s^2 (the sample variance) is an unbiased estimator for σ^2 (the population variance).
- This adjustment corrects for the bias that occurs when using sample data to estimate population variability.

Note: Most statistical software calculates variance and standard deviation using $(n-1)$. Unless otherwise specified, always assume sample data.

Variance: Average Squared Deviation

Step-by-Step Example - Bolt Diameter:

5 measurements: 10.02, 10.04, 10.05, 10.06, 10.08 mm (Mean = 10.05 mm)

X (mm)	$(X - \bar{X})$	$(X - \bar{X})^2$
10.02	-0.03	0.0009
10.04	-0.01	0.0001
10.05	0.00	0.0000
10.06	+0.01	0.0001
10.08	+0.03	0.0009
	Sum = 0	Sum = 0.0020

$$s^2 = \frac{0.0020}{5-1} = \frac{0.0020}{4} = 0.0005 \text{ mm}^2$$

Problem: Variance is in squared units (mm^2) - hard to interpret!

Standard Deviation: The Key Measure

Definition: Square root of variance (converts back to original units)

$$s = \sqrt{s^2} = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}}$$

Continuing Previous Example:

$$s = \sqrt{0.0005} = 0.0224 \text{ mm}$$

Interpretation: On average, bolts deviate $\pm$0.0224 mm from the mean of 10.05 mm.

Coefficient of Variation: Comparing Different Scales

Definition: Standard deviation as percentage of mean

$$CV = \frac{s}{\bar{X}} \times 100\%$$

Why use CV? Allows comparing variability across different measurements/units.

Real Problem:

You manage both:

- **Small components:** diameter mean = 5 mm, $s = 0.1$ mm
- **Large components:** length mean = 500 mm, $s = 5$ mm

Which is more consistent? Can't compare using just s (0.1 vs. 5 - different scales).

Coefficient of Variation: Comparing Different Scales

Solution - Use CV:
Small component: $CV = \frac{0.1}{5} \times 100\% = 2.0\%$
Large component: $CV = \frac{5}{500} \times 100\% = 1.0\%$

Conclusion: Large component is relatively more consistent (1.0% vs. 2.0%)

Use CV for:

- Comparing variability across different units/scales
- Quality benchmarking (industry standards often spec'ify CV limits)
- Multi-product manufacturing lines with different dimensions

Complete Worked Example: Full Analysis

Scenario: Manufacturing tests 12 aluminum rods for hardness **Data:** 85, 90, 88, 92, 87, 91, 89, 86, 93, 88, 90, 89 HV

Step 1: Organize (Tabulation) Ordered: 85, 86, 87, 88, 88, 89, 89, 90, 90, 91, 92, 93

Step 2: Central Tendency

- Mean: $\bar{X} = \frac{1058}{12} = 88.17$ HV
- Median: Position = $\frac{12+1}{2} = 6.5 \rightarrow \frac{89+89}{2} = 89$ HV
- Mode: 88, 89, 90 (each appears twice - trimodal)

Complete Worked Example: Full Analysis

Step 3: Dispersion

X	$(X - 88.17)$	$(X - 88.17)^2$
85	-3.17	10.03
86	-2.17	4.69
87	-1.17	1.37
88	-0.17	0.03
88	-0.17	0.03
89	+0.83	0.69
89	+0.83	0.69
90	+1.83	3.36
90	+1.83	3.36
91	+2.83	8.03
92	+3.83	14.69
93	+4.83	23.36
	Sum = 0	Sum = 70.33

$$s = \sqrt{\frac{70.33}{11}} = \sqrt{6.39} = 2.53 \text{ HV}, \quad CV = \frac{2.53}{88.17} \times 100\% = 2.87\%$$

Final Report: Mean = 88.17 HV, Median = 89 HV, s = 2.53 HV. Variability = 2.87%. Data symmetric (Mean Median). Quality acceptable.

Summary: Four-Step Analysis Process

Step 1: EDIT & PROCESS

- Correct errors, standardize units
- Code responses, classify data

Step 2: TABULATE

- Organize into formal frequency table
- Calculate frequencies, verify totals

Step 3: CENTRAL TENDENCY

- Calculate Mean, Median, Mode
- Identify typical value and check for skewness

Step 4: DISPERSION

- Calculate Range, Standard Deviation, CV
- Assess variability and consistency

Result: Raw numbers $\rightarrow$ Organized data $\rightarrow$ Statistical insight $\rightarrow$ Engineering decision

Key Insight: Without proper processing and analysis, data is worthless. These steps transform numbers into actionable knowledge!

When to Use Each Statistic

Question	Use	Why
What's typical?	Mean	Most common, uses all data
What's middle value?	Median	Robust to outliers
What's most common?	Mode	For qualitative/categorical
How spread out?	Range	Quick rough estimate
How much variation?	Std Dev	Enables predictions, $\pm$intervals
Compare across scales?	CV	Unit-independent, relative
Are data symmetric?	Mean vs Median	If equal $\rightarrow$ symmetric
Quality consistent?	Small Std Dev	Lower = better consistency

Golden Rule: Always report **BOTH mean and standard deviation**

Mean $\pm$ Std Dev tells complete story (location + spread) **Example:** Tensile strength = 475 $\pm$ 15 MPa **This single statement conveys both typical value and consistency**

Complete Analysis Checklist

Final verification before reporting results:

Pre-Analysis:

- ☐ Data properly edited (no errors, outliers investigated)?
- ☐ Coding consistent, classes well-defined?
- ☐ Table has title, units, totals, source?

Central Tendency:

- ☐ Mean calculated correctly (sum/n)?
- ☐ Median found (data ordered, correct position)?
- ☐ Mode identified (highest frequency)?
- ☐ Mean vs Median compared (skewness check)?

Complete Analysis Checklist

Dispersion:

- ☐ Standard deviation calculated (correct formula, n-1)?
- ☐ CV computed (if comparing across scales)?
- ☐ Empirical rule applied (68-95-99.7%)?

Final Check:

- ☐ Results make engineering sense?
- ☐ Proper units and precision reported?
- ☐ Conclusions supported by data?
- ☐ Limitations documented?

Quick Reference: Formula Summary

Central Tendency:

Mean: $\bar{X} = \frac{\sum X}{n}$ **Median:** Middle position $= \frac{n+1}{2}$ **Mode:** Most frequent value

Dispersion:

Range: $R = X_{\max} - X_{\min}$ **Variance:** $s^2 = \frac{\sum(X-\bar{X})^2}{n-1}$

Std Dev: $s = \sqrt{\frac{\sum(X-\bar{X})^2}{n-1}}$ **Coeff. of Variation:** $CV = \frac{s}{\bar{X}} \times 100\%$

Empirical Rule (for normal distributions):

- **68%** of data: Mean $\pm$ 1s
- **95%** of data: Mean $\pm$ 2s
- **99.7%** of data: Mean $\pm$ 3s

Skewness Detection:

- Mean Median $\rightarrow$ Symmetric distribution
- Mean $>$ Median $\rightarrow$ Right skewed (high outliers)
- Mean $<$ Median $\rightarrow$ Left skewed (low outliers)